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MIRZA FARHAN BIN TAREK, University of Delaware, Newark, DE, United States

RAPHAEL POULAIN, University of Delaware, Newark, DE, United States

RAHMATOLLAH BEHESHTI, University of Delaware, Newark, DE, United States

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Fairness-Optimized Synthetic EHR Generation for Arbitrary Downstream Predictive Tasks

Mirza Farhan Bin Tarek
University of Delaware
Newark, DE, USA
mfarhan@udel.edu

Raphael Poulain
University of Delaware
Newark, DE, USA
rpoulain@udel.edu

Rahmatollah Beheshti
University of Delaware
Newark, DE, USA
rbi@udel.edu

Abstract

Among various aspects of ensuring the responsible design of AI tools for healthcare applications, addressing fairness concerns has been a key focus area. Specifically, given the wide spread of electronic health record (EHR) data and their huge potential to inform a wide range of clinical decision support tasks, improving fairness in this category of health AI tools is of key importance. While such a broad problem (mitigating fairness in EHR-based AI models) has been tackled using various methods, task- and model-agnostic methods are rare. In this study, we aimed to target this gap by presenting a new pipeline that generates synthetic EHR data, which is not only consistent with (faithful to) the real EHR data but also can reduce the fairness concerns (defined by the end-user) in the downstream tasks, when combined with the real data. We demonstrate the effectiveness of our proposed pipeline across various downstream tasks and two different EHR datasets. Our proposed pipeline can add a widely applicable and complementary tool to the existing toolbox of methods to address fairness in health AI applications, such as those modifying the design of a downstream model.[1].

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → **Health informatics**.

Keywords

Synthetic Data Generation, Fairness, Machine Learning

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1 Introduction

Integrating ML and AI tools into healthcare has become increasingly feasible with the widespread adoption of large-scale electronic health record (EHR) systems, which contain rich information related to various medical concepts, including diagnoses, procedures, medications, and lab measurements. Different ML models are helping in various aspects of healthcare, for example, clinical predictive modeling [5, 6, 17, 31], subtyping patients [23] and treatment suggestions [34, 35, 41]. These models present both opportunities and challenges in healthcare systems. For example, while ML models can help with more effective diagnosis and treatment, they also risk exacerbating health inequities among historically marginalized groups, if they are not carefully designed[12, 14], or the data used for training has inherent bias [1].

To achieve fairness in clinical ML applications, the literature has presented a diverse array of fairness notions and corresponding methodologies for the user to make the model output fair according to the desired characterization of fairness [22]. Synthetic data generation techniques present a novel opportunity to address fairness considerations at the data level, especially in the case of underrepresented populations. This approach enables the creation of datasets that inherently embody fairness properties, potentially ensuring that subsequent models trained on these data will exhibit fair behavior in arbitrary downstream tasks. The primary objective of fair synthetic data generation is to produce a dataset that closely approximates the original data distribution while simultaneously mitigating or eliminating discriminatory biases present in the source data. It is important to note that this process of fairness-aware synthetic data generation inevitably alters the underlying data distribution. Consequently, models trained on such synthetically generated fair datasets may experience a degree of performance degradation compared to those trained on the original, unmodified data. One alternative option to using only synthetic data or naively adding synthetic data to the existing data (e.g., for oversampling) for fairness mitigation is generating customized synthetic data that especially targets addressing fairness when augmented to (merged with) the existing data.

This study aims to propose an end-to-end pipeline for the above purpose, which is generating on-demand synthetic data to mitigate fairness concerns in arbitrary downstream tasks using EHR data. Our focus is distinct from prior work on synthetic EHR generation[1, 37], as we target not the synthetic data alone, but its combination with existing real data for usage in the user-defined downstream ML tasks. Additionally, we shift away from the common notion of fair data generation[42, 43], as fairness (a sociotechnical phenomenon) is greatly context-dependent, and hence fair synthetic data by itself may have a limited value. In particular, our key contributions are:

(1) We propose an end-to-end pipeline for mitigation of fairness in downstream health ML tasks, with modules for processing EHR data, generating fairness-optimized (FO) synthetic EHR, and a prediction module. The fairness notion, the generator, and the prediction module can be user-defined.

[1]The codebase for our project is available at https://github.com/healthylaife/FairSynth

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(2) Through a series of extensive experiments, we show that our proposed pipeline can effectively help with mitigating fairness in various downstream tasks while maintaining a reasonable trade-off between improving fairness and the performance of the downstream tasks.

2 Related Work

2.1 Fairness in Machine Learning

Fairness within ML can be viewed in various ways. One common way relates to categorizing fairness into two primary types: (i) group fairness [8], and (ii) individual fairness [13]. Our focus in this research is primarily on group fairness, which involves assessing the discrepancy in performance across the groups (including the protected and privileged groups). This emphasis aims to highlight systemic biases against specific groups. As defined in the literature, group fairness exhibits diverse interpretations, often selected based on stakeholders' perspectives and, at times, presenting conflicting behaviors in specific contexts [24, 27].

On a related note, a common (despite its caveats) way to categorize the methods to mitigate bias in the ML community includes (i) pre-processing techniques, which modify input data to eliminate information correlated with sensitive attributes; (ii) in-processing methods, which impose fairness constraints during the model learning process; and (iii) post-processing approaches, which adjust model predictions after training [32]. Our research will process the data and optimize the generator model for fairness.

2.2 Synthetic and Fair EHR Data Generation

Many approaches exist for generating synthetic EHR data. A common approach is using Generative Adversarial Networks (GANs), which are commonly used for generating images and have been used in generating EHR data with moderate success [2, 7, 9, 16, 36, 38, 46]. Generating sequential EHR data is challenging with GANs as they often produce individual outputs without temporal connections. To address this limitation, various strategies have been employed, such as aggregating data into one-time steps [33, 44, 47], creating data representations [9], or combining both approaches [2, 7]. GANs also face difficulties handling high-dimensional and sparse real-world EHR data, restricting existing synthetic EHR GAN approaches to relatively low-dimensional output by aggregating visits and medical codes or removing rare codes. For example, methods generating longitudinal data, such as `LongGAN` [36] and `EHR-M-GAN` [21], focus solely on dense lab time series with dimensions under a hundred. Despite the potential of conditioning GANs on external factors and labels, such as demographics or disease phenotype labels, limited exploration of this capability exists in current works on EHR generation.

Another synthetic data generation approach is diffusion models, which have emerged as an approach for realistic synthetic data generation. Specifically, denoising diffusion probabilistic models (DDPMs) are a class of latent variable models that can learn the underlying distribution of data by transforming samples into standard Gaussian noise and training the model to denoise these corrupted samples back to their original forms [4].

In addition to their stable training, DDPMs have demonstrated superior performance compared to generative adversarial networks (GANs) in generating synthetic images, audio, and tabular data [4]. Recently, He et al. [19] proposed `medDiff`, a DDPM specifically designed for electronic health records (EHRs), showcasing the benefits of diffusion models over existing methods. However, `medDiff` works best to generate continuous values as it employs a Gaussian diffusion process, assuming the data follows a Gaussian distribution, even though medical data often contains both continuous features (e.g., blood test results) and categorical features (e.g., sex and ethnicity). While one can round the outputs of `medDiff` to obtain discrete values, a more principled approach would be to generate categorical data directly, eliminating the need for post-processing and potentially leading to more accurate synthetic samples. Following this, Ceritli et al. [4] proposed `TabDDPM` which is a diffusion model to generate tabular EHRs with continuous or temporal (which changes over time e.g. lab results, vital signs) and static data (which does not change over time e.g. patient gender, ethnicity, etc.). However, the author did not include the effectiveness of these methods in generating datasets that can achieve fair results in downstream tasks.

Finally, with the recent rise of large language models, researchers have already started using them for synthetic EHR data generation. Large language models utilize transformers [40] in their architectures and they have been used for many tasks in different domains for example in developing large language models [10, 30, 39] and in healthcare for primordial prevention [29], few-shot learning [28]. `HALO` or Hierarchical Autoregressive Language Model for synthetic EHR data generator can generate a probability density function over different medical codes, clinical visits, and patient records. This allows the generation of realistic EHR data [37]. In addition to autoregressive methods, GPT or Generative Pre-trained Transformer models have been used recently to generate EHR. In the work by Pang et al. [25], the authors proposed a `CEHR-GPT` framework that treats patient generation as a language modeling problem similar to `HALO` but they have used the state-of-the-art GPT language model for the task. Furthermore, their patient representation can also preserve track of the different visit types and discharge facilities for inpatient visits and maintain the temporal information.

3 Problem Statement

Structured electronic health records (EHRs) are multilevel longitudinal data that show each patient as a series of visits. Each visit is represented by a set of medical codes, including static patient data (e.g. demographics, disease phenotype labels, diagnoses, procedures, and medications codes), and continuous data (e.g. lab test results). In our project, we developed a pipeline for generating Fairness-Optimized (FO) augmented EHR data by implementing a fairness objective F in the synthetic data generator and later joining the FO synthetic data and the real data together to create an augmented dataset. Now, we establish a formal representation below:

Given a real EHR dataset R, a fairness objective function F, and a generation process function G, the goal is to generate a Fairness-Optimized (FO) synthetic EHR dataset R' and augment R. We represent our pipeline in as:

$$R \cup G(R, F) = R \cup R'. \tag{1}$$

The fairness objective F is incorporated as a loss function $\mathcal{L}F$ in the generator's total loss $\mathcal{L}_{model}$, as shown in Eq 2. This objective is

weighted by parameter λ and combines with the standard generator loss $L(y, \hat{y})$. The fairness term implements a group fairness criterion (e.g. demographic parity) that ensures proportionally similar positive outcomes between minority and majority groups across sensitive attributes $S \in [[1, M]]$. Here, S represents categorical demographic variables with M possible values (e.g., gender, age, race, and ethnicity), and the outcome is a binary mortality prediction. This fairness constraint applies to demographic groups maintaining group-level fairness properties.

$$\mathcal{L}_{model} = \mathcal{L}(y, \hat{y}) + \lambda \mathcal{L}_F \tag{2}$$

4 Pipeline Description

The pipeline mainly consists of three parts: i) data pre-processing, ii) FO synthetic data generation, and iii) data augmentation (Figure 1). In the following sections, we describe our pipeline in detail.

4.1 Data pre-processing

Pre-processing can include filtering the data based on specific features and imputing the missing values. Here, we process the input EHR data in a format suitable for the generator G. The data R comprises N patient records r, represented as a series of visits V, over T time steps. We represent the i-th patient record as $r_i = \{V_1^i, V_2^i, V_3^i, \ldots, V_T^i\}$. Each visit V also contains several medical codes $m_1^t, m_2^t, \ldots, m_k^t \in C$, labs $l_1^t, l_2^t, \ldots, l_n^t \in L$ where $k = |V_C^{(t)}|$ and $n = |V_L^{(t)}|$, C is the set of medical codes (diagnosis, procedures and medications) and L is the set of labs. The pipeline allows users to filter the data based on specific criteria and output preprocessed data. The patient records also contain the labels D for the disease phenotypes as well as static information S for each patient, where $S \in \{1, 2, \ldots, K\}^N$, and K is the number of subgroups.

4.2 Fairness-Optimized Synthetic Data Generation

The second component of our pipeline is a synthetic EHR data generator. Even though our pipeline is generator agnostic, we have used the HALO EHR generator (represented as G) to demonstrate our pipeline. HALO uses a two-module approach: visit-level and code-level. The visit-level module processes a patient's medical history using transformer decoder blocks, generating visit history representations that summarize the patient's trajectory. The code-level module generates each variable within a visit based on the patient's history and previously generated variables, ensuring intra-visit cohesion. It takes the concatenated visit history embeddings and the preprocessed data R as input to create a representation of the codes with visit history, fed through masked linear layers to model the distribution of each patient code, preserving the autoregressive property. The final output represents the code probabilities used for generating synthetic data.

4.2.1 Fairness Optimization Objective Function. To ensure that the generator can produce fairness-optimized data, we have implemented a fairness optimization objective function $\mathcal{L}_F$ in the generator's forward pass, ensuring that the code probabilities are not biased toward a certain subgroup.

Let us represent the predicted probabilities for each medical code as P, where $P \in \mathbb{R}^{N \times T \times C}$. Let us also denote M as EHR masks, where $M \in \{y_1, y_2, \ldots\}^{N \times T \times C}$.

For model loss $\mathcal{L}(y, \hat{y})$, we implement the binary cross-entropy (BCE) loss which measures the dissimilarity between the predicted probabilities and the ground truth labels. The general formula for BCE is as follows,

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{M} y_{ij} \log(p_{ij}), \tag{3}$$

where N is the number of rows and M is the number of classes. However, in our project, the binary cross entropy loss function works on patient visits and whether a certain medical code exists or not in a certain visit. We can now rewrite equation 2 as the following:

$$\mathcal{L}_{model} = \mathcal{L}_{BCE} + \lambda \mathcal{L}_F \tag{4}$$

The generalized fairness optimization (FO) objective function, $\mathcal{L}_F$ (Algorithm 1), quantifies the disparity in the predicted probabilities between different subgroups of the sensitive attribute. Based on the fairness task F the user wants to tackle, it takes the predicted code probabilities P and the sensitive attribute labels S as inputs.

Algorithm 1 Fairness Optimization (FO) Objective

Require: $P \in \mathbb{R}^{N \times C}$ - Predicted probabilities for each outcome, where N is the number of instances and C is the number of possible outcomes
Require: $S \in 1, 2, ..., K^N$ - Sensitive attribute labels for each instance, where K is the number of subgroups
Require: F is fairness objective
Ensure: $\mathcal{L}_F$ - Fairness loss

```
S ← unique(S) {Get the unique sensitive attribute subgroups}
Initialize fairness metric-specific variables
for i ← 1 to N do
    s ← S_i {Get the sensitive attribute subgroup of instance i}
    j ← index(s, 𝒮) {Find the index of the subgroup in 𝒮}
    Update fairness metric-specific variables based on Pi and j
end for
Calculate fairness metric-specific values
L_F ← Compute fairness loss based on the fairness metric
return L_F
```

Algorithm 1 takes two inputs: the predicted probabilities for each possible outcome (P) and the sensitive attribute labels for each instance (S). The goal is to compute a fairness loss $\mathcal{L}_F$ that measures the disparity in the model's predictions across different subgroups of the sensitive attribute. The user can specify the fairness objective based on what type of bias the user plans to address.

The algorithm starts by identifying the unique subgroups S_K present in the sensitive attribute labels. It then initializes the fairness metric-specific variables that will be used to calculate the fairness loss. These variables may include subgroup-wise statistics, such as sums of predicted probabilities or counts of instances.

Next, the algorithm iterates over the instance is, from 1 to N. For each instance, it retrieves the sensitive attribute subgroup s and

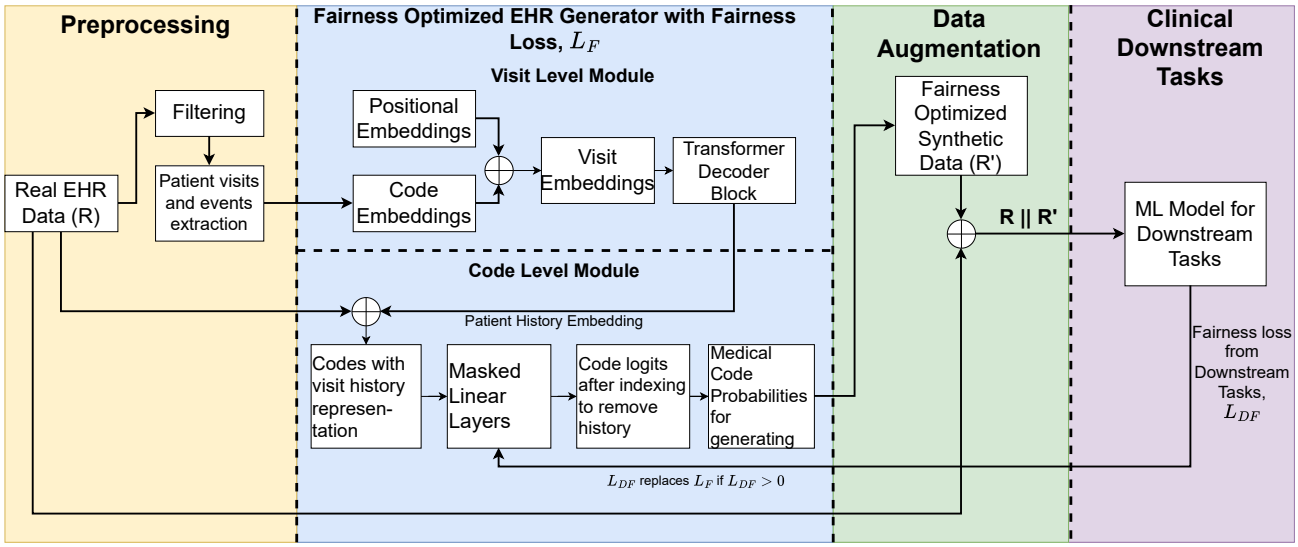


Figure 1: The diagram of our proposed pipeline consists of three major components: data pre-processing, data generation, and data augmentation. The pre-processing stage prepares the EHR data for the generator model alongside any necessary filtering. The generator model generates fair synthetic EHR data. In our project, we have used the HALO model [37] with fairness optimization. It has a visit-level module that processes visit embeddings using masked self-attention and a feed-forward network. The output of the self-attention layer and the feed-forward network are normalized (circular blocks) before inputting to the next stage. The code-level module generates each variable within a visit based on the patient's history and the previously generated variables in the same visit, ensuring maximum intra-visit cohesion (meaning that the generated data should represent a coherent visit where the variables are similar to a typical visit). The loss function of the generator incorporates the proposed fairness objective function or $\mathcal{L}_F$. After calculating a fairness metric F from the downstream task, we feed it back into the loss function as $\mathcal{L}_{DF}$.

finds the corresponding index j in the set of unique subgroups $\mathcal{S}$. The fairness metric-specific variables are then updated based on the predicted code probabilities P_i, for instance, i and its subgroup index j. This step accumulates the necessary information to compute the fairness metric. After processing all cases, the algorithm calculates the fairness metric-specific values using the accumulated information. Finally, the fairness loss $\mathcal{L}_F$ is calculated based on the fairness metric-specific values which can also be viewed as regularization terms in the model's overall loss function during training.

We can also input the fairness performance metric of a downstream prediction task (for example, disparate impact in mortality prediction task) into the model loss calculation. We call this value $\mathcal{L}_{DF}$. If $\mathcal{L}_{DF}$ has been calculated after the downstream prediction task, then we say $\mathcal{L}_{DF} > 0$ to represent the existence of $\mathcal{L}_{DF}$ and calculate the model loss as the sum of $\mathcal{L}_{BCE}$ and $\mathcal{L}_{DF}$. This ensures that the model is optimized directly based on the fairness performance in the specific downstream task of interest. Combining both $\mathcal{L}_F$ and $\mathcal{L}_{DF}$ in the calculation of model loss $\mathcal{L}_{model}$, we represent the final equation for the loss calculation in equation 5.

$$\mathcal{L}_{model} = \begin{cases} \mathcal{L}_{BCE} + \lambda \mathcal{L}_{DF}, \mathcal{L}_{DF} \mathcal{L}_{DF} > 0 \\ \mathcal{L}_{BCE} + \lambda \mathcal{L}_{F}, otherwise \end{cases} \tag{5}$$

The fairness weight λ controls the trade-off between the model's predictive performance (measured by the BCE loss) and its fairness to the sensitive attribute (measured by the generalized fairness loss).

During training, the model's parameters are updated to minimize the total loss $\mathcal{L}$. This encourages the model to learn to make accurate predictions (by minimizing the BCE loss) while simultaneously reducing the disparity between sensitive attribute subgroups (by minimizing the generalized fairness loss). The goal is to find a balance between predictive performance and fairness, as determined by the fairness weight λ.

4.3 Augmentation of Real Data

After we generate the FO synthetic data R' and ensure that the real EHR data R and the FO synthetic data are in the same format, we join both datasets to finally obtain $R \cup R'$. This merged pool of real and synthetic data is used in the downstream prediction task.

5 Experiments

5.1 Data Description

To demonstrate the efficacy of our pipeline, we have performed a set of experiments over two datasets: the MIMIC-III dataset [20] and the PIC (Pediatric Intensive Care) dataset [45]. Both of them are available in the Physionet data bank [15]. The MIMIC-III dataset

is sourced from the Medical Information Mart for Intensive Care (MIMIC), to analyze inpatient records. This extensive dataset encompasses anonymized EHRs originating from patients admitted to the Beth Israel Deaconess Medical Center in Boston, MA, USA, from 2001 to 2012. The data contains information for each patient regarding hospital stay, including laboratory measurements, medications administered, and vital signs. From the whole dataset, we have used the ICU stay dataset. It contains 46,520 patients with 25 disease phenotype labels which were defined by the MIMIC benchmark [18]. Among the patients, 32,950 patients identify as male and the remaining ones identify as females. The dataset has more than 30 ethnicities, however, to simplify processing, we decided to group the different ethnicities into 5 categories based on the ethnicity names containing White, Black, Hispanic, or Asian keywords. So, the groups are White, Black, Hispanic, Asian, and Others. The White ethnicity has the most samples in the dataset, which is 41,325 samples, and the rest belong to other groups.

On the other hand, the Pediatric Intensive Care (PIC) database is a large pediatric-specific, single-center, bilingual database comprising information relating to children admitted to critical care units at the Children's Hospital, Zhejiang University School of Medicine. The dataset has a similar structure to the MIMIC-III dataset, however, the dataset is smaller, having 13,449 patients: 7,728 of whom identify as male and the rest identify as females. Considering the ethnic distribution, it is very skewed toward the Han ethnic group (13,300 patients and the rest of the patient population belongs to the Miao, Tujia, Buyei, Yi, and Hui ethnic groups.

5.2 Data Extraction and Feature Selection

MIMIC-III dataset: We worked with the PATIENTS, ADMISSIONS, DIAGNOSES_ICD, PRESCRIPTIONS, and PROCEDURE_ICD tables. For each patient, we extracted the time series events, episode-level information (patient age, gender, ethnicity), and patient mortality following the approach explained by Harutyunyan et al. [18]. The dataset contains static data (e.g., gender, ethnicity, diagnosis codes, procedure codes, expiry flag of the patient) and continuous data (e.g., lab results, age). Our cohort from MIMIC-III consists of ICU patients.

PIC dataset: From the PIC dataset, we worked with the PATIENTS, ADMISSIONS, and DIAGNOSES_ICD tables. This dataset only includes children admitted to critical care units and contains static data (e.g., gender, ethnicity, diagnosis codes, and the patient's expiry flag) and continuous data (e.g., lab results, and age).

In both cases, the continuous data were converted to discrete values using granular discretization, and static data were converted into labels. Finally, for both cohorts, our downstream task is to predict mortality in a patient's next visit based on the current visit.

5.3 Experimental Methods

We conducted several experiments to explore the efficacy of our proposed pipeline in various scenarios. In our experiments, we have used 'ethnicity' as the sensitive attribute and kept aside other sensitive attributes for future experiments. In our pipeline, fairness optimization is facilitated through an FO objective, allowing users to select the specific fairness metric (DI/WTPR). Throughout our experiments, we utilize DI or disparate impact as the FO objective function (implementation can be found in appendix A.1). DI quantifies the adverse impact of a model's predictions on a protected group by comparing selection rates between disadvantaged and advantaged groups [3]. The ideal value of DI is 1 or near 1. Subsequently, in addition to DI, we assess the fairness of the downstream model using the worst true positive rate (worst TPR or WTPR) which denotes the lowest TPR among all protected groups and underscores the model's capability to identify positive cases within the most disadvantaged group [11]. The higher this is, the better the model is identifying true positives for the subgroups.

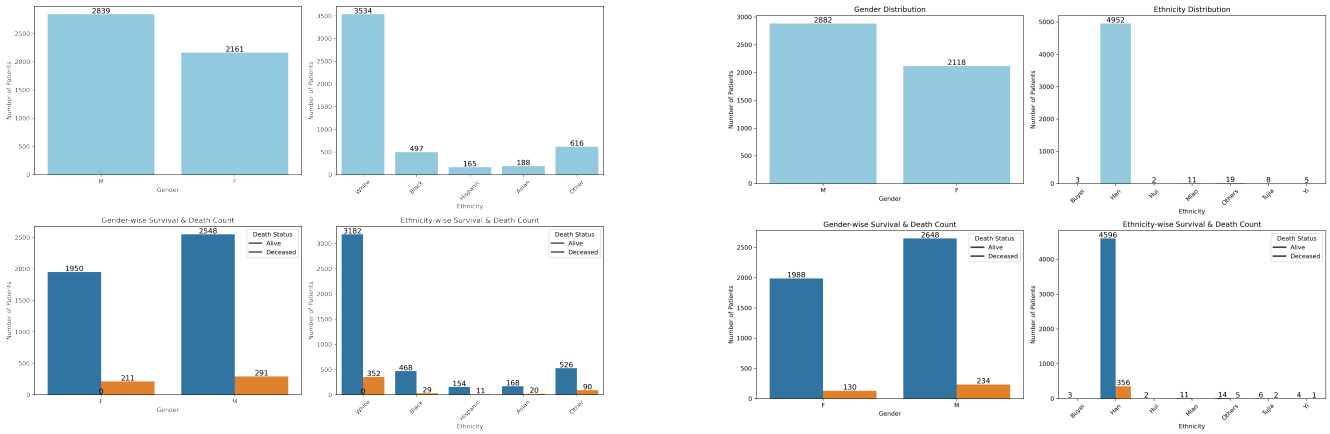
Downstream Prediction Model: Both of these metrics were evaluated on two separate transformer-based prediction models that were trained on multiple subsets of the real dataset and the synthetic dataset. The model's embedding layer holds the embeddings for input features: visit codes, disease labels, ethnicity, and gender. The final output is a single value denoting whether a patient will be alive in the next hospital visit. The linear layers include separate linear transformations for visit codes, disease labels, ethnicity, and gender. These linear layers project the input features into a common 128-dimensional space.

The core of the model is the transformer component, which contains separate transformer layers for each input feature (visit codes, disease labels, ethnicity, and gender). Each transformer layer consists of a transformer block that includes a multi-headed attention module for capturing dependencies within the input sequence, and a position-wise feed-forward module for non-linear transformations. The transformer layers also include layer normalization and dropout regularization. The outputs from the transformer layers are concatenated and passed through a final fully connected layer to produce the model's output, which is a single value. The model is trained using an Adam optimizer with a learning rate of 0.001, no weight decay, and no maximum gradient norm. The batch size is set to 64, and the model is trained for 30 epochs.

The training data were built in two ways: i) augmenting a varying number of real data samples with a fixed number of synthetic samples and ii) augmenting a fixed number of real data samples with a varying number of synthetic samples. The whole dataset was divided into 80-10-10 train-validation-test split.

Baselines: We compare our pipeline (*Real+FairSynth*) to three other comparable approaches to work with the downstream prediction model. Using these approaches, we created eight curated datasets, six of which are related to the baseline datasets. Among the eight datasets, four of them include patients from the MIMIC-III dataset and the rest include patients from the PIC dataset. The four approaches are as follows:

- *Real-Only*, where only the data from the real dataset is used to train the downstream models.
- *Real+Synth*, where the real data is augmented with synthetic data generated through the synthetic generator without implementing the FO part.
- *Real+Oversample*, where the real data is over-sampled using the popular double prioritized bias correction method Pias et al. [26]. The double prioritized (DP) bias correction method addresses data imbalance in a specific subgroup, such as a particular ethnicity, by incrementally replicating data points of the minority iteratively. In our experiments, we replicated



(a) MIMIC-III Cohort

(b) PIC Cohort

Figure 2: Distribution of different genders and ethnicities in the MIMIC-III and the PIC Cohort.

patient data belonging to each minority ethnicity so that the distribution is less skewed.

- *Real+FairSynth (Our Method)*, where the real data is augmented with the synthetic data that is generated using our proposed pipeline.

Experiment types To study the impact of the fairness objective function in fair EHR data generation, we first observe the fairness metrics in the real data. Table 1 compares DI and WTPR for the 3 baselines and our proposed method for both datasets. In our experiments, we aim to evaluate the impact of the size of the real dataset and the synthetic data set on the fairness and accuracy of the prediction model for fairness weight λ=1.2.

5.4 Fairness Evaluation in Mortality Prediction with MIMIC-III Dataset

5.4.1 Experiment 1. Our proposed method did not achieve the best results for the DI metric however for the worst-case TPR metric, where a higher value is better, our proposed method achieves better results for 2,500 and 5,000 samples with a TPR of 0.75±0.05 at 2,500 samples and 0.78±0.07 at 5,000 samples which are higher than the baselines.

5.4.2 Experiment 2. Experiments were conducted where 2,500 real data samples were augmented with different sizes of synthetic samples. In the lower half of table 1a, we can observe that increasing synthetic samples yields improvements for all methods. However, the proposed Real+FairSynth achieves DI values nearest to the ideal value of 1. At 500 synthetic samples, Real+FairSynth has a DI of 0.98±0.22. For a higher number of synthetic data, the values are closer to 1, however, the baselines achieve better performance in this case.

Analysis of the worst-case True Positive Rate (WTPR) indicates that the Real+FairSynth method shows increasing effectiveness as the synthetic sample size grows. With 500 synthetic samples, Real+FairSynth yields a WTPR of 0.24±0.14, exceeding baseline performance. This difference becomes more pronounced at 1,000 synthetic samples, where Real+FairSynth achieves 0.72±0.02, compared to 0.19±0.03 and 0.15±0.01 for alternative approaches. At 2,000 samples, Real+FairSynth reaches a WTPR of 0.83±0.06, while Real+Oversample and Real+Synth maintain lower rates of 0.25 and 0.15, respectively.

These findings suggest that augmenting a fixed real dataset with synthetic samples can enhance fairness metrics and that the proposed Real+FairSynth method is particularly effective.

5.5 Fairness Evaluation in Mortality Prediction with PIC Dataset

5.5.1 Experiment 1. The PIC dataset experiments followed a similar structure to MIMIC-III, examining two scenarios: varying real sample sizes with fixed synthetic data, and fixed real sample sizes with varying synthetic data. As shown in Table 1b, the Real+FairSynth method yields higher DI values across configurations. At both 1,000 and 2,500 data samples, the DI measurements approach 1, surpassing the baseline methods. Regarding worst-case True Positive Rate (WTPR), Real+FairSynth demonstrates enhanced performance for 1000 real sample sizes (0.29 ± 0.09), compared to Real-Only (0.12 ± 0.03), Real+Oversample (0.08 ± 0.03), and Real+Synth (0.13 ± 0).

5.5.2 Experiment 2. We can observe from the lower half of Table 1b that our proposed method achieved improved results for a variable number of synthetic data samples.

To summarize, the comparative analysis (results are shown in tables 1a and 1b provides evidence that our proposed methodology outperforms the baselines on both key fairness metrics of DI and WTPR across different ranges of sample sizes evaluated. This also shows that, even with a few real samples, adding synthetic samples generated by our proposed method can improve the fairness metrics.

5.6 Prediction Performance

We have also observed the prediction performance of the transformer-based model alongside the fairness metrics for the baseline datasets

Sample Size		Disparate Impact (closer to 1 is better)				WTPR↑				F1-Score↑			
Real	Synth	R	R+Over	R+Synth	R+FairSynth	R	R+Over	R+Synth	R+FairSynth	R	R+Over	R+Synth	R+FairSynth
1,000	2,500	.98(.02)	**.99(.03)**	.97(.04)	.39(.66)	**.48(.07)**	.26(.01)	.17(.03)	.44(.23)	**.53(.06)**	.27(.09)	.25(.03)	.45(.02)
2,500	2,500	.98(.05)	**.98(.04)**	.97(.04)	1.20(.20)	.51(.06)	.24(.07)	.17(.01)	**.75(.05)**	**.51(.09)**	.28(.07)	.27(.06)	.49(.01)
5,000	2,500	**.99(.02)**	.99(.02)	.98(.03)	1.10(.09)	.67(.06)	.14(.02)	.14(.02)	**.78(.07)**	**.55(.03)**	.28(.06)	.27(.05)	.46(.02)
2,500	500	-	.98(.03)	.98(.04)	**.98(.22)**	-	.16(.04)	.16(.02)	**.24(.14)**	-	.27(.03)	.25(.05)	**.48(.04)**
2,500	1,000	-	**.99(.03)**	.98(.02)	1.28(.42)	-	.19(.03)	.15(.01)	**.72(.02)**	-	.29(.03)	.26(.04)	**.50(.03)**
2,500	2,000	-	1.02(.04)	**.99(.06)**	1.03(.44)	-	.25(.01)	.15(0)	**.83(.06)**	-	.33(.02)	.26(.04)	**.49(.01)**

(a) MIMIC-III Dataset Results

Sample Size		Disparate Impact (closer to 1 is better)				WTPR↑				F1-Score↑			
Real	Synth	R	R+Over	R+Synth	R+FairSynth	R	R+Over	R+Synth	R+FairSynth	R	R+Over	R+Synth	R+FairSynth
1,000	2,500	.87(.51)	1.41(.81)	.42(0)	**.66(.23)**	.12(.03)	.08(.03)	.13(0)	**.29(.09)**	.38(.25)	**.62(.02)**	.14(.12)	.03(.04)
2,500	2,500	.98(.05)	.39(.49)	1.38(.32)	**.85(.15)**	.06(.05)	.1(.04)	**.27(.16)**	.06(.05)	.36(.25)	**.6(.05)**	.12(.06)	.10(.07)
5,000	2,500	.76(.77)	**.95(.18)**	1.11(.76)	.80(.14)	.05(.02)	.15(.13)	**.19(.17)**	.15(.20)	.35(.25)	**.57(.01)**	.09(.05)	.07(.06)
2,500	500	-	1.34(.44)	1.3(.52)	**.6(0)**	-	.06(.03)	.1(.6)	**.03(.03)**	-	**.56(.07)**	.51(.17)	.06(.06)
2,500	1,000	-	**.99(.03)**	.98(.02)	1.28(.42)	-	.13(.11)	.04(.01)	**.02(.04)**	-	**.57(.06)**	.45(.02)	.04(.09)
2,500	2,000	-	1.02(.04)	**.99(.06)**	1.03(.44)	-	.05(.02)	.02(0)	**.13(.16)**	-	**.58(.05)**	.41(.02)	.06(.08)

(b) PIC Dataset Results

Table 1: Combined Results for Real and Synthetic Samples from MIMIC-III (top) and PIC (bottom) datasets. Mean (SD) values are shown. R: Real, Over: Over Sample. Here the bold numbers are the values closest to ideal values. Mean(standard deviation). ↑: Higher is better

and the Real+FairSynth. The prediction performance is measured using the F1-score. From tables 1a and 1b, we can observe the f1-scores for the real data and augmented datasets for different numbers of real and synthetic samples for both datasets. It can be observed that, for both MIMIC-III and PIC datasets, augmenting the data with our proposed fairness-optimized synthetic data does not increase the predictive performance compared to the other dataset, which showcases the trade-off between prediction performance and fairness.

5.7 Trade-off Between Fairness and Prediction Accuracy

In this study, we investigate the trade-off between prediction performance and fairness metrics by varying the value of the fairness weight parameter, λ. This analysis is crucial in understanding how different values of λ impact both the F1-scores and fairness metrics such as Disparate Impact (DI) and Weighted True Positive Rate (WTPR).

5.7.1 Impact on Prediction Performance. Table 2a illustrates the F1-scores for different values of λ on the MIMIC-III dataset across three different sample sizes (1,000, 2,500, and 5,000). As λ increases from 0.5 to 1.5, there is a noticeable decline in the F1 scores. For example, for 5,000 samples, the F1-score reduces from 0.46 ± 0.02 at λ = 0.5 to 0.24 ± 0.02 at λ = 1.5. This trend clearly shows that higher values of λ negatively impact the prediction performance, as the declining F1 scores indicate.

5.7.2 Impact on Fairness Metrics. Conversely, Table 2b, 2c, 2e and 2f present the DI and WTPR values for the same variations in λ and sample sizes. Here, a higher value of λ generally enhances the fairness metrics. However, we observe that increasing the λ too much, for example, from 1.2 to 1.5, affects the DI impact positively but WTPR negatively, indicating that changing the lambda value causes variation in the fairness metrics.

In summary, we have observed a trade-off between accuracy (F1-scores) and fairness (DI and WTPR) metrics is a crucial aspect of model optimization. Lower values of λ prioritize prediction performance, resulting in higher F1 scores but potentially less fair outcomes. In contrast, higher values of λ improve fairness metrics at the expense of prediction accuracy.

This trade-off suggests that the specific goals of the application should guide the choice of λ. If the priority is to maximize prediction accuracy, lower values of λ may be preferable. However, if fairness is a critical objective, higher values of λ should be considered despite the associated decrease in prediction performance. Balancing this trade-off is key to developing models that are both effective and fair, tailored to the ethical and operational requirements of the specific use case. From our experiments, we can safely say that λ = 1.2 provides a good trade-off between the fairness and prediction performance for both the MIMIC-III and PIC datasets.

5.8 Limitations

Our proposed pipeline has several limitations. Firstly, while our pipeline in the current version is limited to tabular data, it still establishes a valuable foundation for generating FO synthetic data and has shown good results. Including other modalities like medical images or medical notes in future versions might improve the pipeline's output.

# of Real Samples	F1-scores for MIMIC-III↑			
	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 1.2$	$\lambda = 1.5$
1,000	**0.55(0.03)**	0.1(0.09)	0.45(0.02)	0.22(0.08)
2,500	**0.52(0.02)**	0.13(0.09)	0.49(0.01)	0.24(0.01)
5,000	**0.46(0.02)**	0.13(0.03)	0.45(0.03)	0.24(0.02)

(a) F1-scores for different λ

# of Real Samples	DI scores for MIMIC-III			
	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 1.2$	$\lambda = 1.5$
1,000	1.28(1.21)	1(0.05)	0.89(0.66)	**1.00(0.06)**
2,500	0.99(0.22)	1.28(0.42)	1.20(0.28)	**0.99(0.02)**
5,000	1.10(0.)	0.99(0.16)	1.10(0.09)	**0.99(0.04)**

(b) DI scores for different λ

# of Real Samples	WTPR scores for MIMIC-III↑			
	λ = **0.5**	λ = **1**	λ = **1.2**	λ = **1.5**
1,000	0.64(0.24)	**0.72(0.03)**	0.44(0.23)	0.15(0.08)
2,500	0.74(0.09)	0.72(0.02)	**0.75(0.05)**	0.15(0.01)
5,000	0.78(0.07)	0.73(0.08)	**0.78(0.07)**	0.15(0.02)

(c) WTPR for different λ values

# of Real Samples	F1-scores for PIC↑			
	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 1.2$	$\lambda = 1.5$
1,000	0.04(0.08)	**0.1(0.09)**	0.03(0.04)	0.07(0.10)
2,500	0.05(0.00)	**0.13(0.09)**	0.10(0.07)	0.02(0.05)
5,000	0.06(0.05)	**0.13(0.03)**	0.07(0.06)	0.04(0.06)

(d) F1-scores for different λ

# of Real Samples	DI scores for PIC			
	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 1.2$	$\lambda = 1.5$
1,000	1.28(1.21)	1(0.05)	0.89(0.66)	**1.00(0.06)**
2,500	0.99(0.22)	1.28(0.42)	1.20(0.28)	**0.99(0.02)**
5,000	1.10(0.09)	0.99(0.16)	1.10(0.09)	**0.99(0.04)**

(e) DI scores for different λ

# of Real Samples	WTPR scores for PIC↑			
	λ = **0.5**	λ = **1**	λ = **1.2**	λ = **1.5**
1,000	0.64(0.24)	**0.72(0.03)**	0.44(0.23)	0.15(0.08)
2,500	0.74(0.09)	0.72(0.02)	**0.75(0.05)**	0.15(0.01)
5,000	0.78(0.07)	0.73(0.08)	**0.78(0.07)**	0.15(0.02)

(f) WTPR for different λ values

Table 2: Effect of changing λ values on F1-score and fairness metrics on the MIMIC-III (2a, 2b, 2c) and PIC dataset (2d, 2e, 2f) for variable number of real samples and 2,500 synthetic samples generated by our proposed method. We can observe that we get a higher f1-score for lower λ values showing better prediction performance by trading off fairness. Mean(standard deviation). ↑: Higher is better.

Furthermore, we did not train the generator and downstream prediction models on very large-scale EHRs. We expect that our model would maintain its performance on larger datasets and running the models for a longer time would improve the output quality even further. Finally, our current pipeline version supports augmentation as the sole debiasing method. Other types of fairness mitigation methods (like inference-time methods) can be merged with our pipeline.

6 Conclusion

In this study, we presented a new pipeline for mitigating bias in clinical downstream tasks with the help of fairness-optimized (FO) augmented EHR data. Specifically, we used the FO objective function to ensure that combining the real data with the generated synthetic data can improve the fairness performance of downstream predictive models running on the data. Through a series of comprehensive experiments on MIMIC-III and pediatric ICU datasets, we showed that adding FO objectives in the generator and augmenting real data with FO synthetic data can reduce the biases observed for different sensitive attributes in downstream mortality prediction tasks. Even though our method solely focuses on data-related methods where we try to improve fairness by augmenting, combining our proposed method with model-related techniques (where the downstream model is modified) and inference-related techniques (where post-processing is performed on the output) is possible.

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A Appendix

A.1 Description of the FO Objective Functions Used in Experiments

Disparate impact examines the likelihood of receiving a positive classification. However, instead of focusing on the difference between unprivileged and privileged groups, it considers the ratio of these probabilities. The concept of disparate impact has roots in legal fairness assessments for selection processes, where the 80% rule is sometimes applied to determine whether a process exhibits disparate impact (ratio below 0.8) or not [3]. We can formalize DI as follows:

Let SR_p be the selection rate for the protected group, and SR_n be the selection rate for the non-protected group. The selection rate for a group is calculated as the number of favorable outcomes (e.g., positive predictions, approvals) divided by the total number of instances in that group.

So, the disparate impact ratio is DI then calculated as:

$$DI = \frac{SR_p}{SR_n} \tag{6}$$

The implementation is shown in algorithm 2.

Algorithm 2 Disparate Impact Loss Function

Require: $P \in \mathbb{R}^{N \times C}$ - Predicted probabilities for each code, where N is the number of patients and C is the number of codes
Require: $G \in 0...N^N$ - {Subgroup labels for a sensitive attribute}
Ensure: $\mathcal{L}_{DI}$ - Disparate impact loss

```
G ← unique(G) {Get the unique subgroup labels}
S_g ← 0^|𝒢| {Initialize sensitive attribute-wise positive prediction sums}
N_g ← 0^|𝒢| {Initialize sensitive attribute-wise patient counts}
for i ← 1 to N do
    g ← G_i {Get the sensitive attribute value of patient i}
    j ← index(g, 𝒢) {Find the index of the sensitive attribute in 𝒢}
    p_i ← Σ_{c=1}^{C} 1(P_{i,c} > 0.5) {Count positive predictions for patient i}
    S_{g_j} ← S_{g_j} + p_i {Add positive prediction count to the corresponding value of the sensitive attribute}
    N_{g_j} ← N_{g_j} + 1 {Increment the count for the corresponding sub group}
end for
R_g ← S_g / N_g, ∀g ∈ 𝒢 {Calculate the positive prediction rate for each sub group}
DI ← R_{g_0} / R_{g_1} {Compute the disparate impact ratio}
L_DI ← |1 − DI| {Calculate the disparate impact loss}
return L_DI
```

The loss function inputs the predicted probabilities for each code (P) and the sensitive attribute labels for each patient (G). It first identifies the unique sensitive attribute labels present in the dataset ($\mathcal{G}$). Then, it initializes two arrays, S_g and N_g, to store the sum of positive predictions and the count of patients for each sensitive attribute group, respectively.

The algorithm then iterates over each patient i and determines their subgroup g. It finds the corresponding index j of the sensitive attribute in the unique label $\mathcal{G}$. For each patient, the function counts the number of positive predictions (p_i) by summing the indicator function $\mathbf{1}(P_{i,c} > 0.5)$ over all codes c. This indicator function returns 1 if the predicted probability $P_{i,c}$ is greater than 0.5 (i.e., a positive prediction) and 0 otherwise. The count of positive predictions is added to the corresponding sensitive attribute's sum S_{g_j}, and the patient count for that sensitive attribute N_{g_j} is incremented.

After processing all patients, each subgroup's positive prediction rate R_g is calculated by dividing the sum of positive predictions S_g by the patient count N_g for each gender g. The disparate impact ratio DI is then computed as the ratio of the positive prediction rates between the subgroups.

Finally, the disparate impact loss $\mathcal{L}_{DI}$ is calculated as the absolute difference between 1 and the disparate impact ratio DI. This loss function quantifies the deviation from perfect equality (i.e., a disparate impact ratio of 1) between the gender groups. A higher loss value indicates a larger disparate impact, while a lower loss value suggests a more balanced and fair model.

Worst performing TPR (the lowest) or the Worst TPR. It is a useful fairness metric because it focuses on the model's performance for the most disadvantaged subgroup. It helps identify if there are significant disparities in the model's performance across different subgroups defined by the protected attribute. Suppose $A \in [[1, N]]$, where A is a categorical sensitive attribute with N possible values(e.g. gender, race, ethnicity, religion), $Y \in [0, 1]$ and $\hat{Y} \in [0, 1]$ as the ground truth and the prediction of a binary predictor, respectively.

$$\text{Worst-Case TPR} = \min_{n \in E} Pr(\hat{Y} = 1 | A = n, Y = 1) \tag{7}$$

A smaller value (closer to zero) is ideal for the parity-based measures, but for the worst TPR, a greater value(closer to one) is desired.

A.2 Experimental Setup

We split the data into an 80-10-10 ratio for training, validation, and testing. Due to limited computing resources and GPU power, we ran the synthetic EHR generator using the Adam optimizer with a batch size of 10, a sample batch size of 25, and a learning rate of 10^{-4}.

All of our experiments were run using the built-in ML models, specifically a transformer-based model, from the PyHealth library. These models utilize static data, such as demographic information like gender and ethnicity, and continuous time-series data.

The experiments were performed on three environments: Two Windows PCs and in the cloud using Google Colab. The first machine had an Intel Core i9-9900K processor, 32 GB of RAM, and an NVIDIA GeForce RTX 2080Ti GPU with 12GB memory. The second machine featured an Intel Core i9-13900HX processor, 32 GB of RAM, and an NVIDIA GeForce RTX 4070 GPU with 8GB memory. In Google Colab, we utilized the T4, L4, and A100 GPUs. All the

codes were run using Python version 3.11.5, CUDA version 12.1, and PyTorch 2.1.2.